

SaNaPower

Power analysis and sample size for the F tests: a step-by-step manual

User manual • version 0.1.0 • October 2026

SaNaPower • Power analysis • F tests • 0.1.0 [Manual \(PDF\)](#)

▼ What is this screen? What do I do here?
Power analysis for the F tests: multiple regression (R^2 and R^2 increase) and ANOVA or ANCOVA (one-way, or one effect of a factorial design). Work out how many participants you need before collecting data (a priori), the power of a planned or finished study (post hoc), or the smallest effect a fixed sample can detect (sensitivity). No data file is needed and nothing leaves your device. The sentence at the bottom can go straight into your proposal or methods section.

1 - Which analysis will you run?

- ☒ **Multiple linear regression, F-test of R^2 (R^2 deviation from zero)**
Does the set of predictors explain any variance? Effect size $f^2 = R^2 / (1 - R^2)$.
- ☐ **Multiple linear regression, F-test of the R^2 increase (hierarchical regression, a block of predictors, a moderator)**
Does adding the tested predictors raise R^2 beyond the others? Effect size $f^2 = (R^2 \text{ with } - R^2 \text{ without}) / (1 - R^2 \text{ without})$.
- ☐ **One-way ANOVA, fixed effects (Compare three or more groups)**
One-way ANOVA, effect size f .
- ☐ **ANOVA or ANCOVA, one effect of a factorial design (main effect or interaction)**
Two-way or larger designs, with or without covariates.

2 - What do you want to find?

- ☒ **A priori power analysis (The sample size I need)**
Most common in proposals.
- ☐ **Post hoc power analysis (The power of a given sample size)**
For a fixed number of respondents.
- ☐ **Sensitivity power analysis (The smallest effect I can detect)**
Useful when the sample size is already fixed.

Effect size f^2
0.15
Conventions: small 0.02, medium 0.15, large 0.35 (Cohen, 1988). Better: take it from earlier studies.

Significance level α
0.05

Power ($1 - \beta$)
0.80
Plan for a power of .80 at $\alpha = .05$ unless the field expects otherwise.

Number of predictors
4

N = 85
How to report it: An a priori power analysis for a multiple regression with 4 predictors ($H_0: R^2 = 0$) indicated that 85 participants are needed to detect $f^2 = 0.15$ with $\alpha = .05$ and power = .80 (achieved power .803).

► References

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Worked examples: regression, hierarchical regression, ANOVA and ANCOVA

Prof. S. Sabraz Nawaz and Prof. Ghazanfar Ali

SaNaSoft • 2026

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About this manual

SaNaPower answers two questions that every proposal, ethics form and thesis asks: how many participants do I need, and how likely is my study to find an effect that is really there? It covers the F-test family: multiple linear regression (the test of R^2 , and the test of the R^2 increase) and the analysis of variance (a one-way ANOVA, and one effect of a factorial ANOVA or ANCOVA). It runs in your web browser. There is nothing to install, no account to create and no data file to upload, and nothing you type leaves your computer. Once the page has been opened, it also works without an internet connection.

This manual shows each of the four designs with a worked example, tells you which effect size to enter and where to find it, and gives the sentence to paste into your methods section. The screenshots and numbers come from real runs of SaNaPower 0.1.0.

Note: what SaNaPower does not cover

SaNaPower contains the four F-test designs of this manual and nothing else. It has no t-tests, correlation, chi-square, repeated-measures or multivariate (MANOVA, Hotelling's T^2) power analysis. For the broader family of analyses, SaNaML (statistics and machine learning, same authors) has a Power analysis screen with the t-test, correlation and chi-square designs as well.

Using SaNaPower

1. Getting started

Open the SaNaPower page in any current browser (Chrome, Edge, Firefox, Safari). The screen has two numbered cards and a result box (Figure 1).

1. 1 • Which analysis will you run? Choose one of the four designs (Chapters 3 to 6).
2. 2 • What do you want to find? Choose the kind of power analysis (Chapter 2).
3. Type the effect size, the significance level α , the power and the details of the design. The result and a ready-made sentence appear at once; there is no button to press.

The screenshot shows the SaNaPower web application interface. At the top, there is a header with the logo, the title "SaNaPower", the subtitle "Power analysis · F tests · 0.1.0", and a link to the "Manual (PDF)". Below the header, there is a section titled "What is this screen? What do I do here?" which provides a brief explanation of the application's purpose. The main content area is divided into two numbered sections. Section 1, "Which analysis will you run?", contains four radio button options: "Multiple linear regression, F-test of R² (R² deviation from zero)", "Multiple linear regression, F-test of the R² increase (hierarchical regression, a block of predictors, a moderator)", "One-way ANOVA, fixed effects (Compare three or more groups)", and "ANOVA or ANCOVA, one effect of a factorial design (main effect or interaction)". Section 2, "What do you want to find?", contains three radio button options: "A priori power analysis (The sample size I need)", "Post hoc power analysis (The power of a given sample size)", and "Sensitivity power analysis (The smallest effect I can detect)". Below these sections, there are input fields for "Effect size f²", "Significance level α ", and "Power (1 - β)". The "Effect size f²" field is set to 0.15, the "Significance level α " field is set to 0.05, and the "Power (1 - β)" field is set to 0.8. There is also a "Number of predictors" field set to 4. Below these fields, a large green box displays the result: "N = 85". At the bottom, there is a "References" section and a copyright notice.

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- ☐ **Post hoc power analysis (The power of a given sample size)**
For a fixed number of respondents.
- ☐ **Sensitivity power analysis (The smallest effect I can detect)**
Useful when the sample size is already fixed.

Effect size f^2
Conventions: small 0.02, medium 0.15, large 0.35 (Cohen, 1988).
Better: take it from earlier studies.

Significance level α

Power (1 - β)
Plan for a power of .80 at $\alpha = .05$ unless the field expects otherwise.

Number of predictors

N = 85

How to report it: An a priori power analysis for a multiple regression with 4 predictors ($H_0: R^2 = 0$) indicated that 85 participants are needed to detect $f^2 = 0.15$ with $\alpha = .05$ and power = .80 (achieved power .803).

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Figure 1. SaNaPower as it opens: a regression with four predictors and a medium effect ($f^2 = 0.15$) needs $N = 85$.

Concept: errors, power and effect size

A **Type I error** is a false positive (finding an effect that is not there); its probability is the significance level α , usually .05. A **Type II error** is a false negative (missing an effect that is there); its probability is β . **Power** = $1 - \beta$ is the probability of finding an effect that really exists; .80 is the usual target. Power rises with the sample size, the size of the true effect and α . The **effect size** is the smallest effect that would matter to you, or one taken from earlier studies; the conventions of Cohen (1988), small, medium and large, are a fallback, not a substitute for thinking.

2. The three kinds of power analysis

A priori power analysis (The sample size I need) is done before data collection: you give the effect size, α and the power you want, and SaNaPower returns the smallest sample size that reaches that power. This is what proposals and ethics committees ask for.

Post hoc power analysis (The power of a given sample size) gives the power of a sample size you already have (or can afford) for an assumed effect size. Use it to plan. Do not use the effect size observed in the same data to explain a non-significant result after the fact: that power is a function of the p-value and adds no information; the confidence interval of the effect is the better description (Hoenig & Heisey, 2001).

Sensitivity power analysis (The smallest effect I can detect) gives the smallest effect that a fixed sample can detect with the power you choose. With $N = 120$, power .80, $\alpha = .05$ and four predictors, a regression can detect $f^2 = 0.104$ or larger (Figure 2). Sensitivity analysis is the honest answer when the sample is fixed, for example by a population of 120 employees.

SaNaPower Power analysis · F tests · 0.1.0 [Manual \(PDF\)](#)

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1 · Which analysis will you run?

- ☒ **Multiple linear regression, F-test of R^2 (R^2 deviation from zero)**
Does the set of predictors explain any variance? Effect size $f^2 = R^2 / (1 - R^2)$.
- ☐ **Multiple linear regression, F-test of the R^2 increase (hierarchical regression, a block of predictors, a moderator)**
Does adding the tested predictors raise R^2 beyond the others? Effect size $f^2 = (R^2 \text{ with} - R^2 \text{ without}) / (1 - R^2 \text{ without})$.
- ☐ **One-way ANOVA, fixed effects (Compare three or more groups)**
One-way ANOVA, effect size f .
- ☐ **ANOVA or ANCOVA, one effect of a factorial design (main effect or interaction)**
Two-way or larger designs, with or without covariates.

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Most common in proposals.
- ☐ **Post hoc power analysis (The power of a given sample size)**
For a fixed number of respondents.
- ☒ **Sensitivity power analysis (The smallest effect I can detect)**
Useful when the sample size is already fixed.

Significance level α

Power ($1 - \beta$)
Plan for a power of .80 at $\alpha = .05$ unless the field expects otherwise.

Total sample size N

Number of predictors

$f^2 = 0.104$

How to report it: With $N = 120$ and power = .80, a multiple regression with 4 predictors ($H_0: R^2 = 0$) can detect effects of $f^2 = 0.104$ or larger at $\alpha = .05$ (sensitivity analysis).

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Figure 2. Sensitivity analysis: with $N = 120$ the regression can detect $f^2 = 0.104$ or larger.

Tip: plan for drop-out and missing data

The numbers SaNaPower gives are the sample you need *after* exclusions. If you expect 10% of respondents to drop out, give unusable answers or leave variables blank, recruit the target divided by .90.

3. Multiple regression: the F-test of R^2

Choose **Multiple linear regression, F-test of R^2 (R^2 deviation from zero)**. The question is whether the whole set of predictors explains any variance in the outcome ($H_0: R^2 = 0$). Enter the **number of predictors** and the effect size $f^2 = R^2 / (1 - R^2)$. Cohen's conventions are small 0.02, medium 0.15 and large 0.35.

With four predictors and a medium effect ($f^2 = 0.15$), $\alpha = .05$ and power .80, SaNaPower reports $N = 85$ (achieved power .803), as in Figure 1. If an earlier study found $R^2 = .172$ with four predictors, $f^2 = .172 / .828 = 0.21$ and the sample size falls to $N = 62$. With only $N = 60$ available, the power for $f^2 = 0.15$ is .621.

How to report it

"An a priori power analysis for a multiple regression with 4 predictors ($H_0: R^2 = 0$) indicated that 85 participants are needed to detect $f^2 = 0.15$ with $\alpha = .05$ and power = .80 (achieved power .803)."

Tip: partial models and PLS-SEM

The number of predictors is the number of predictors in the model you will test. For a structural model with several outcomes, use the outcome that has the most predictors, because it is the most demanding.

4. Multiple regression: the F-test of the R^2 increase

Choose **Multiple linear regression, F-test of the R^2 increase (hierarchical regression, a block of predictors, a moderator)**. The question is whether the predictors you add raise R^2 beyond the predictors already in the model: a block in a hierarchical regression, or an interaction term that tests a moderator.

Enter the number of predictors you **test** (the block you add) and the **total number of predictors** in the final model, including every control variable. The effect size is $f^2 = (R^2 \text{ with} - R^2 \text{ without}) / (1 - R^2 \text{ without})$. If you know the two R^2 values rather than f^2 , open **I know the R^2 values, not f^2** , type them, and click **Use this f^2** (Figure 3).

Take three control variables and two tested predictors that raise R^2 from .30 to .40. Then $f^2 = (.40 - .30) / (1 - .40) = 0.167$. With $\alpha = .05$ and power .90, SaNaPower reports $N = 80$ (achieved power .903), the value published for the same inputs (Faul et al., 2009).

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 Most common in proposals.

☐ Post hoc power analysis (The power of a given sample size)
 For a fixed number of respondents.

☐ Sensitivity power analysis (The smallest effect I can detect)
 Useful when the sample size is already fixed.

Effect size f^2

 Conventions: small 0.02, medium 0.15, large 0.35 (Cohen, 1988). Better: take it from earlier studies.

Significance level α

Power ($1 - \beta$)

 Plan for a power of .80 at $\alpha = .05$ unless the field expects otherwise.

Predictors tested (the block you add)

Total predictors in the model

 Including the tested ones and every control variable.

▼ I know the R^2 values, not f^2

R^2 without the tested predictors

R^2 with all predictors

$f^2 = (.40 - .30) / (1 - .40) = 0.167$ Use this f^2

N = 80

How to report it: An a priori power analysis for a test of the R^2 increase from 2 of 5 predictors in a multiple regression indicated that 80 participants are needed to detect $f^2 = 0.17$ with $\alpha = .05$ and power = .90 (achieved power .903).

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Figure 3. The R^2 increase: 2 of 5 predictors tested, $f^2 = 0.167$ from $R^2 = .30$ and .40, power .90 needs $N = 80$.

How to report it

"An a priori power analysis for a test of the R^2 increase from 2 of 5 predictors in a multiple regression indicated that 80 participants are needed to detect $f^2 = 0.17$ with $\alpha = .05$ and power = .90 (achieved power .903)."

5. One-way ANOVA

Choose **One-way ANOVA, fixed effects (Compare three or more groups)**. Enter the **number of groups** and Cohen's f : small 0.10, medium 0.25 and large 0.40. If you know η^2 from an earlier study, $f = \sqrt{(\eta^2 / (1 - \eta^2))}$; for $\eta^2 = .054$, $f = \sqrt{(.054 / .946)} = 0.24$.

With three groups, $f = 0.25$, $\alpha = .05$ and power $.80$, SaNaPower reports $N = 159$, which is 53 per group (achieved power $.805$) (Figure 4). The total is always a multiple of the number of groups, because the groups are equal in size. A large effect ($f = 0.40$) needs only $N = 66$ (22 per group).

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One-way ANOVA, effect size f .
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- ☒ A priori power analysis (The sample size I need)
Most common in proposals.
- ☐ Post hoc power analysis (The power of a given sample size)
For a fixed number of respondents.
- ☐ Sensitivity power analysis (The smallest effect I can detect)
Useful when the sample size is already fixed.

Effect size Cohen's f Significance level α Power ($1 - \beta$)

0.25 0.05 0.8

Conventions: small 0.10, medium 0.25, large 0.40 (Cohen, 1988). Better: take it from earlier studies.

Plan for a power of .80 at $\alpha = .05$ unless the field expects otherwise.

Number of groups

3

N = 159

How to report it: An a priori power analysis for a one-way ANOVA with 3 groups indicated that 159 participants (53 per group) are needed to detect Cohen's $f = 0.25$ with $\alpha = .05$ and power = $.80$ (achieved power $.805$).

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Figure 4. One-way ANOVA with three groups: $f = 0.25$ needs 53 participants per group, $N = 159$.

How to report it

"An a priori power analysis for a one-way ANOVA with 3 groups indicated that 159 participants (53 per group) are needed to detect Cohen's $f = 0.25$ with $\alpha = .05$ and power = $.80$ (achieved power $.805$)."

6. One effect of an ANOVA or ANCOVA

Choose **ANOVA or ANCOVA, one effect of a factorial design (main effect or interaction)**. Use it for two-way and larger designs, with or without covariates, when the hypothesis concerns one effect (a main effect or an interaction). Enter three numbers.

- **Degrees of freedom of the effect:** for a main effect, its number of levels – 1; for an interaction, the product of (levels – 1) of its factors. In a 2×2 design the interaction has 1 df; in a 2×3 design it has $1 \times 2 = 2$ df.
- **Number of groups (cells of the whole design):** for a 2×3 design, 6; for a 2×2 design, 4.
- **Number of covariates:** 0 for an ANOVA. The effect size f is the one *after* adjusting for the covariates.

For a 2×2 design with the pre-test as one covariate, the interaction (1 df, 4 groups, 1 covariate) with $f = 0.25$, $\alpha = .05$ and power .80 needs $N = 128$ (achieved power .801) (Figure 5).

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Does adding the tested predictors raise R^2 beyond the others? Effect size $f^2 = (R^2 \text{ with} - R^2 \text{ without}) / (1 - R^2 \text{ with})$.
- ☐ One-way ANOVA, fixed effects (Compare three or more groups)
One-way ANOVA, effect size f .
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- ☐ Post hoc power analysis (The power of a given sample size)
For a fixed number of respondents.
- ☐ Sensitivity power analysis (The smallest effect I can detect)
Useful when the sample size is already fixed.

Effect size Cohen's f 0.25 Conventions: small 0.10, medium 0.25, large 0.40 (Cohen, 1988). Better: take it from earlier studies.	Significance level α 0.05	Power ($1 - \beta$) 0.8 Plan for a power of .80 at $\alpha = .05$ unless the field expects otherwise.
Degrees of freedom of the effect 1 A main effect: its levels – 1. An interaction: the product of (levels – 1) of its factors.	Number of groups (cells of the whole design) 4 For a 2×3 design: 6.	Number of covariates 1 0 for an ANOVA. The effect size f is the one after adjusting for the covariates.

N = 128

How to report it: An a priori power analysis for an ANCOVA effect test (1 df, 4 groups, 1 covariate) indicated that 128 participants are needed to detect Cohen's $f = 0.25$ with $\alpha = .05$ and power = .80 (achieved power .801).

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Figure 5. One effect of an ANCOVA: 1 df, 4 groups, 1 covariate, $f = 0.25$, power .80 needs $N = 128$.

How to report it

"An a priori power analysis for an ANCOVA effect test (1 df, 4 groups, 1 covariate) indicated that 128 participants are needed to detect Cohen's $f = 0.25$ with $\alpha = .05$ and power = .80 (achieved power .801)."

Note: the effect size is for the effect, not the design

f belongs to the one effect you test. The interaction of a 2×2 design is usually smaller than its main effects, so plan from the interaction when it is the hypothesis.

7. How the results were checked

SaNaPower uses the noncentral F distribution with noncentrality $\lambda = f^2 \times N$ (for the ANOVA designs, f^2 is Cohen's f squared) and the degrees of freedom of each test: ($u, N - u - 1$) for the R^2 test with u predictors; (tested, $N - \text{total} - 1$) for the R^2 increase; and (df of the effect, $N - \text{groups} - \text{covariates}$) for the ANOVA and ANCOVA effect. Results were compared with published worked examples (Faul et al., 2009) and with independent calculations in SciPy. For example: R^2 increase, $f^2 = 0.167$, 2 of 5 predictors, power .90: $N = 80$, power .903; one-way ANOVA, $f = 0.204$, 3 groups, power .80: $N = 237$; ANOVA effect, $f = 0.707$, 2 df, 36 groups, $N = 108$: power above .9999.

Exercise: plan three studies

(a) A hierarchical regression adds a block of three predictors to four controls and expects R^2 to rise from .25 to .31. How many participants are needed at $\alpha = .05$ and power .80? (b) You will compare four teaching methods with a one-way ANOVA and expect a medium effect ($f = 0.25$). How many participants per group? (c) A 2×3 experiment has two covariates and you plan to test the interaction with $f = 0.20$. How many participants for power .80? Answers: (a) $f^2 = (.31 - .25) / (1 - .31) = 0.087$, $N = 130$ (3 tested of 7 predictors); (b) 45 per group, $N = 180$; (c) 2 df, 6 groups, 2 covariates: $N = 244$. Write the reporting sentence for each.

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